Machine_Learning_Engineer — Step 4

Build Machine Learning Projects

Detailed Learning Notes • CareerCompass

1. Learning Objective

Learn the complete ML project lifecycle from problem framing and data collection to evaluation, documentation and reproducible delivery.

2. Core Concepts — Detailed Breakdown

Problem framing

Translate a vague requirement into a measurable prediction or decision problem. Define target, prediction horizon, constraints, success metrics and what action will follow a prediction.

Data pipeline

Collect, validate, clean and document data. Check duplicates, missing values, inconsistent categories, outliers and leakage. Record how each transformation changes the data.

Feature engineering

Create useful variables from raw data while respecting the time boundary of the prediction. Encode categorical data, scale numerical data where appropriate, and avoid features that would only be known after the prediction.

Experimentation

Track dataset versions, preprocessing choices, hyperparameters, metrics and random seeds. A model result without a reproducible experiment is difficult to trust.

Error analysis

Do not stop at a score. Inspect false positives, false negatives, subgroup performance and failure cases. Error analysis often reveals data-quality or problem-definition issues.

Communication

A project should explain the problem, data, method, evaluation, limitations and next steps. A portfolio project is stronger when another person can reproduce it.

3. Recommended Learning Workflow

- Write a one-page problem statement.
- Create a data dictionary before modeling.
- Build a baseline and then iterate.
- Track experiments and save the final preprocessing/model artifact.
- Perform error analysis and document limitations.
- Package the result with a README and reproducible instructions.

5. Hands-on Project / Practice

Build a customer-churn or loan-risk style project using a public dataset. Include data validation, exploratory analysis, a preprocessing pipeline, baseline model, two improved models, metric comparison, error analysis, a model card-style limitations section and a small inference script.

6. Common Mistakes & Pitfalls

- Data leakage through target-derived features.
- Randomly splitting time-dependent data.
- Changing the test set repeatedly until the score looks good.
- Reporting only one metric.
- Leaving notebooks undocumented or impossible to reproduce.

7. Completion Checklist

- I can frame an ML problem clearly.
- I can create a data dictionary and validation checks.
- I can build preprocessing/model pipelines.
- I can compare models fairly.
- I can explain limitations and reproduce my final result.

8. Interview / Assessment Questions

- What makes a good ML problem statement?
- What is data leakage?
- Why should a test set be protected?
- What belongs in a model card or project README?
- How do you perform error analysis?

9. Recommended References

Scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

Machine Learning in 2024 — freeCodeCamp: https://www.youtube.com/watch?v=bmmQA8A-yUA

Kaggle Learn: https://www.kaggle.com/learn

Python for Data Science course: https://www.youtube.com/watch?v=CMEWVn1uZpQ

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.